

8th semester

Analytical Chemistry Sp-I

(2019)

SHORT QUESTIONS

Q1. What is the effect of temperature on conductance of solution ?

The conductance of a solution generally increases with an increase in temperature. This is due to the fact that higher temperatures provide more thermal energy to the ions or molecules in the solution, causing them to move more rapidly and increase their mobility. As a result, the ions or molecules can carry electrical charge more effectively through the solution, leading to higher conductance.

However, it's essential to note that this effect can vary depending on the specific type of solute and solvent involved in the solution.

Q2. How sputtering takes place in glow discharge?

Sputtering in a glow discharge, also known as glow discharge sputtering, is a process in which material is removed from a target surface when bombarded by energetic ions.

A glow discharge is created in a low-pressure gas environment, typically in a vacuum chamber. This discharge generates a plasma consisting of electrons, ions, and neutral gas atoms.

When the high-energy ions collide with the target material, they transfer their kinetic energy to the target atoms. This energy can be enough to overcome the binding forces holding the target atoms in place, causing them to be sputtered off the target surface.

Q3. Give the difference between arc and spark.

Properties	Arc	Spark
Nature of Discharge	An arc is a continuous electrical discharge that occurs between two electrodes. It is characterized by a sustained plasma channel and can persist for an extended period.	A spark is a brief and intermittent electrical discharge that typically occurs as a momentary, high-energy event.
Duration	Arcs can be relatively long-lasting, potentially lasting for seconds to minutes.	Sparks are short-lived and last for a very brief moment, typically milliseconds or less.
Temperature and Energy	Arcs are generally hotter and can reach	Sparks are high-energy discharges but

	higher temperatures than sparks.	are typically cooler than arcs.
Applications	Arcs are commonly used in applications like welding, plasma cutting, and electric arc furnaces.	Sparks are often seen in applications like spark plugs, electrical discharges in air (lightning), and ignition systems.

Q4. Distinguish between specific conductance and molar conductance.

Specific Conductance (κ)	Molar Conductance (Λ_m)
Specific conductance is a measure of the ability of a solution to conduct electrical current.	Molar conductance is a measure of the ability of a solution to conduct electrical current.
It is often denoted as κ (kappa).	It is denoted as Λ_m (lambda m).
Its unit of measurement is Siemens per meter (S/m).	Its unit of measurement is Siemens per mole per meter ($S \cdot m^2/mol$)
Specific conductance measures the conductance of a specific volume of a solution.	Molar conductance measures the conductance per mole of solute.

Specific conductance depends on the concentration of ions in the solution and the temperature.	Molar conductance is particularly useful for comparing the conductivity of solutions with different concentrations because it normalizes for the amount of solute.
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Q5. How oxygen interferes in polarographic analysis?

In polarographic analysis, oxygen typically acts as a depolarizer at the cathode. It consumes electrons produced by the reduction of analytes, like metal ions, at the cathode. This facilitates the flow of current, allowing for the quantitative measurement of the analyte's concentration based on the observed current response. Oxygen's presence at the cathode ensures that a stable reduction potential is maintained during the analysis, aiding in accurate measurements.

Q6. What do you know about residual current in polarography?

In polarography, residual current, also known as residual current noise or background current, refers to the small electrical current that is observed in the absence of a redox reaction occurring at the working electrode. This background current arises from various sources, including electronic noise, impurities in the electrolyte, and other uncontrollable factors that can influence the measured electrical current. Understanding and minimizing residual currents are

important in polarography to obtain accurate and reliable analytical results.

Q7. What are the Advantages and Disadvantages of conductometric titration?

It has following advantages and disadvantages:

Advantages of Conductometric Titration

Sensitivity: Conductometric titration is sensitive to changes in ion concentration and can detect very low concentrations of analytes.

Wide Applicability: It can be used for a wide range of chemical reactions, including acid-base, redox, and precipitation reactions.

Speed: Conductometric titration is relatively quick and can provide results in real-time, allowing for rapid analysis.

Direct Measurement: It provides a direct measurement of the change in ion concentration as the titration.

Disadvantages of Conductometric Titration

Calibration and Standardization: Proper calibration and standardization of the equipment are essential for accurate results, and errors in calibration can lead to inaccuracies.

Complex Sample Preparation: Sample preparation can be more complex for conductometric titration.

Limited Information: Conductometric titration provides limited information about the chemical species involved in the reaction.

Interference from Electrolytes: High concentrations of electrolytes in the sample can lead to interference and inaccuracies in the measurement.

Q8. Give the characteristics of supporting electrolyte in polarography.

Some key characteristics of a supporting electrolyte in polarography include:

Inertness: The supporting electrolyte should be chemically inert and not participate in the electrochemical reaction being studied.

High Ionic Strength: A high concentration of ions in the supporting electrolyte is desirable to provide a stable and conductive environment.

Stability: The supporting electrolyte should remain stable over the course of the experiment.

Compatibility with Electrodes: The supporting electrolyte should also be compatible with the material of the electrodes used.

Solubility: The chosen electrolyte should be highly soluble in the solvent or medium used for the polarographic analysis.

Q9. Write down the basic principle of glow discharge technique.

The glow discharge technique is an analytical method used in various fields, including materials science and surface analysis. Its principle is based on creating a low-pressure plasma discharge in a gas environment to generate a glow or plasma, which interacts with a solid sample.

By analyzing the emitted species and monitoring the discharge conditions, researchers can gain valuable insights into the composition and depth profile of the sample, making it a valuable tool in materials characterization and surface analysis.

Q10. What are the limitations of Amperometry?

Amperometry has limitations, including

- Sensitivity to interference from other electroactive species in the sample, which can lead to inaccurate results.
- Requires a stable and controlled experimental setup, making it less suitable for field measurements.
- Amperometry is typically a slow technique, and it may not provide real-time data for rapid processes.
- Electrode fouling or passivation can occur, affecting the electrode's performance over time.

LONG QUESTIONS

Q.3 (a) Write down the principle or and instrumentation of electrical arcs.

Principle of electrical arcs

The principle of electrical arcs involves the formation of a sustained electrical discharge, known as an arc, between two conductive electrodes in a gaseous or vacuum environment. This phenomenon is driven by the ionization of the gas or vapor between the electrodes due to a high voltage potential.

Instrumentation of electrical arcs

The instrumentation of electrical arcs typically involves a combination of equipment and sensors to control, monitor, and analyze the arc's characteristics. The specific instrumentation used can vary depending on the application, but here are some common components and instruments involved in the instrumentation of electrical arcs:

- ***Power Supply:*** A high-voltage power supply is used to provide the necessary voltage potential to initiate and maintain the electrical arc.
- ***Electrodes:*** Conductive electrodes, typically made of materials like tungsten, copper, or carbon, are used to create the electrical path where the arc forms.
- ***Arc Starters:*** In some applications, devices such as igniters or starting mechanisms are used to initiate the arc.
- ***Gas Supply:*** Depending on the application, a gas supply system may be used to control the gas or vapor environment in which the arc operates.

- ***Arc Controller:*** A control system may be used to regulate the voltage, current, and other operating parameters of the arc. This can include closed-loop control for arc welding and cutting.
- ***Arc Sensors:*** Sensors are employed to monitor various aspects of the arc, including voltage, current, arc length, and arc temperature. Optical sensors can be used to measure the light emission of the arc, providing information about its characteristics.
- ***Spectroscopy Instruments:*** Spectroscopic instruments, such as spectrometers, are used to analyze the spectral lines of the arc's light emission, allowing for chemical composition analysis and temperature measurement.
- ***Cameras:*** High-speed cameras and imaging systems are often used to capture visual images and videos of the arc, helping with analysis and diagnostics.
- ***Safety Equipment:*** Safety measures, such as protective gear for operators and shielding to contain the arc, are essential for applications like welding and cutting.
- ***Data Acquisition Systems:*** Systems for data acquisition and logging are used to collect and store data from various sensors for further analysis and quality control.
- ***Process Control:*** In industrial applications, the instrumentation may be integrated into a broader process control system to ensure consistency and quality in the production process.

The choice of instruments and sensors may vary based on the specific requirements and objectives of the application.

Q.3 (b) What is meant by diffusion current? Discuss factors affecting it.

Definition

Diffusion current refers to the flow of charged particles, such as ions or electrons, due to their random motion caused by thermal energy. In the context of chemistry, it's often associated with the movement of ions in an electrolyte or electrons in a semiconductor.

Factors affecting diffusion current

Factors affecting diffusion current include:

- **Concentration Gradient:** The greater the difference in concentration of the charged particles between two points, the higher the diffusion current. A steeper concentration gradient leads to faster diffusion.
- **Temperature:** Higher temperatures increase the kinetic energy of particles, leading to more rapid diffusion and, consequently, a higher diffusion current.
- **Diffusion Coefficient:** This property depends on the nature of the solute and solvent. Particles with higher diffusion coefficients will diffuse more quickly, resulting in a higher diffusion current.

- ***Membrane or Medium Permeability:*** The type of membrane or medium through which diffusion occurs can significantly affect the diffusion current. Some materials are more permeable to certain ions or electrons than others.
- ***Surface Area and Distance:*** A larger surface area for diffusion and a shorter distance for particles to travel will increase the diffusion current.
- ***Charge of Particles:*** The magnitude and sign of the charge on the particles involved affect their mobility and, consequently, the diffusion current.

Q.4 (a) Explain conductometric titration involving

(i) Strong acid with Strong base

Conductometric titration involving a strong acid with a strong base is a common and straightforward type of titration in analytical chemistry. In this titration, a known concentration of a strong base (NaOH) is gradually added to a solution of a strong acid (HCl). The purpose is to determine the concentration of the acid in the solution.

There are following steps involved in this conductometric titration:

Apparatus Setup

- A burette containing the strong base (NaOH) is set up to allow precise and gradual addition of the titrant to the solution of the strong acid (HCl).
- The solution of the strong acid is placed in a conical flask.

Measurement of Initial Conductance

- The initial conductance (or conductivity) of the strong acid solution is measured using a conductance cell or sensor. This value represents the conductance of the solution before any titrant is added.

Titration

- The strong base (NaOH) is slowly added to the strong acid solution. As the base is added, it reacts with the acid to form water and a salt.
- The addition of the base gradually neutralizes the acidic solution.

Monitoring Conductance

- The conductance of the solution is continuously monitored during the titration. As the acid is neutralized, the number of ions in the solution decreases, leading to a decrease in conductance.

Endpoint Detection

- The titration is continued until a significant and sustained change in conductance is observed. This change indicates that

the stoichiometric equivalence point (neutralization point) has been reached.

Calculation

- Using the volume of the base (NaOH) required to reach the endpoint and the known concentration of the base, the concentration of the strong acid (HCl) can be calculated using the stoichiometry of the reaction.

Repeatability

- It's important to perform the titration at least three times to ensure reproducibility and accuracy in the determination of the acid concentration.

(ii) Weak acid with Strong base

Conductometric titration involving a weak acid with a strong base is a common analytical method used to determine the concentration of a weak acid solution. The strong base, typically NaOH, is gradually added to the weak acid solution. There are following steps involved in this conductometric titration:

Apparatus Setup

- A burette containing the strong base (NaOH) is set up to allow precise and gradual addition of the titrant to the weak acid solution.
- The solution of the weak acid is placed in a conical flask.

Measurement of Initial Conductance

- The initial conductance (or conductivity) of the weak acid solution is measured using a conductance cell or sensor. This value represents the conductance of the solution before any titrant is added.

Titration

- The strong base (NaOH) is slowly added to the weak acid solution. As the base is added, it reacts with the weak acid to form water and the conjugate base of the weak acid.
- The addition of the base gradually neutralizes the weak acid, leading to an increase in the pH of the solution.

Monitoring Conductance

- The conductance of the solution is continuously monitored during the titration. As the weak acid is neutralized, the concentration of ions in the solution changes, leading to changes in conductance.

Endpoint Detection

- The titration is continued until a significant and sustained change in conductance is observed. This change indicates that the stoichiometric equivalence point (neutralization point) has been reached. At this point, the weak acid has been completely neutralized.

Calculation

- Using the volume of the base (NaOH) required to reach the endpoint and the known concentration of the base, the concentration of the weak acid can be calculated using the stoichiometry of the reaction. The pH of the solution at the endpoint can also be determined.

Repeatability

- As with any titration, it's essential to perform the titration at least three times to ensure reproducibility and accuracy in the determination of the weak acid concentration.

Q.4 (b) Discuss the applications of amperometry.

It has various applications in the field of chemistry, including:

Electro analysis

Amperometry is commonly used for quantitative analysis of electroactive species in solution. By measuring the current response at a specific potential, you can determine the concentration of the analyte.

Biosensors

Amperometric biosensors are widely used in medical and biotechnological applications. They can detect specific biological molecules (e.g., glucose, DNA) or other bio-recognition elements.

Water Quality Monitoring

Amperometry is valuable for monitoring water quality. It can detect and quantify contaminants like heavy metals, chlorine, and organic compounds in water samples.

Fuel Cells

Amperometric techniques are used in the development and optimization of fuel cells. They help researchers study the kinetics of electrochemical reactions, assess the performance of fuel cell materials, and improve fuel cell efficiency.

Neuroscience

In neuroscience research, amperometry is employed for real-time monitoring of neurotransmitter release at the cellular level. It enables the study of neuronal activity and the understanding of neurotransmission processes.

Environmental Monitoring

Amperometry plays a role in environmental science by detecting and quantifying pollutants in air and soil samples. It aids in assessing air quality, measuring volatile organic compounds (VOCs), and studying redox reactions in soils.

Pharmaceuticals

In the pharmaceutical industry, amperometry is used for drug analysis and quality control. It can help determine the concentration of active pharmaceutical ingredients (APIs) and identify impurities.

Q.5 (a) Discuss anodic stripping voltammetry.

Quantitative applications of anodic stripping voltammetry (ASV) in chemistry are quite significant. ASV is an electro-analytical technique used to determine the concentration of trace metals and other electroactive species in a solution. It's quantitative applications are:

Trace Metal Analysis

ASV is widely employed for the determination of trace metals like lead, copper, cadmium, and mercury in various samples, including environmental, biological, and industrial samples. It offers high sensitivity, making it suitable for detecting low concentrations of these metals.

Calibration Curves

To quantitatively determine the concentration of a specific metal ion, a calibration curve is constructed by measuring the current response at different known concentrations of the analyte. This curve helps establish a linear relationship between current and concentration.

Detection Limits

ASV can achieve extremely low detection limits, making it useful for

analyzing samples with very low metal ion concentrations, such as in environmental monitoring for compliance with regulatory limits.

Speciation Studies

ASV can also be used to distinguish between different oxidation states or species of a metal ion in a sample, providing valuable information for speciation studies in environmental and biological chemistry.

Monitoring Electroplating Baths

In industrial applications, ASV is used to monitor and control the metal content in electroplating baths, ensuring the quality of plated products.

Pharmaceuticals

ASV is employed in pharmaceutical analysis for the determination of trace metals in drug formulations, helping to ensure product quality and compliance with regulatory standards.

Research and Quality Control

ASV is a valuable tool in both research laboratories and quality control settings for quantitative analysis of trace metals due to its precision and accuracy.

Q.5 (b) Discuss various types of electrodes in polarography.

Polarography is a technique used in electrochemistry to study the behaviour of substances in solution. Various types of electrodes are employed in polarography to facilitate the measurement of current as

a function of applied voltage. Some common types of electrodes used in polarography are:

Dropping Mercury Electrode (DME)

This is one of the most widely used electrodes in polarography. Mercury forms a droplet at the end of a capillary tube, and the potential is applied to this droplet. It offers excellent reproducibility and stability.

Static Mercury Drop Electrode (SMDE)

Unlike DME, in SMDE, the mercury droplet remains stationary. It provides a stable surface for measurements and is often used for trace analysis.

Rotating Disk Electrode (RDE)

RDEs are employed to enhance mass transport of analytes to the electrode surface. The electrode is rotated, creating a dynamic situation that improves sensitivity.

Static Solid Electrode

Solid electrodes made of materials like platinum, gold, or glassy carbon are also used. They are less commonly employed but offer advantages in certain applications.

Gas-Diffusion Electrode

These electrodes are used when studying gases or volatile compounds. They allow gases to diffuse to the electrode surface, making it suitable for gas analysis.

Saturated Calomel Electrode (SCE)

SCE is a reference electrode commonly used in polarography to establish a stable reference potential.

Ag/AgCl Reference Electrode

Another type of reference electrode used in polarography. It consists of silver and silver chloride and is easy to handle.

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(2020)

SHORT QUESTIONS

Q1. What is the effect of dilution on specific conductance. Give reason.

Dilution of a solution typically decreases its specific conductance because it reduces the concentration of ions in the solution, resulting in a lower ability to carry electrical current

Q2. What is the basic principle of anodic stripping voltammetry.

Anodic Stripping Voltammetry (ASV) is an electro-analytical

technique used for the quantitative determination of trace metals, particularly heavy metals, in solution.

First, the analyte is electrodeposited on the working electrode during the deposition step. The oxidation of the analyte occurs on the working electrode during the stripping step, when the peak current is measured.

Q3. What is Ilkovich equation? Give its significance.

The Ilkovic equation, also known as Ilkovic's law or Ilkovic's equation, is a mathematical formula used in electrochemistry to describe the relationship between the current (I), time (t), and the amount of substance (m) that is electroplated or deposited onto an electrode during an electrochemical reaction.

The Ilkovic equation is expressed as:

$$id = knD^{1/3}m^{2/3}t^{1/6}c$$

Significance of Ilkovic equation:

- The Ilkovic equation, named after Croatian chemist Vladimir Ilkovic, is significant in electrochemistry and is used to determine the number of electrons transferred in an electrode reaction.
- It's particularly useful in quantitative analysis and understanding the stoichiometry of electrochemical reactions.
- The equation relates the charge passed during an electrochemical reaction to the number of moles of a substance

involved in the reaction, which can be essential in various applications, such as corrosion studies, battery research, and electroplating processes.

Q4. Give application of conductometry in precipitation titrations.

- Conductometry, the measurement of electrical conductivity in solutions, is commonly used in precipitation titrations for determining the endpoint of a titration.
- Conductometric titrations can be easily automated, which helps reduce human error and increases precision in endpoint detection.
- Conductometric titrations are versatile and can be used for a wide range of precipitate-forming reactions. They are especially common in the determination of halides, sulfides, and various other analytes.
- Conductometric titrations are relatively simple to perform and can provide rapid results.
- Conductometric titrations can detect very low concentrations of ions, making them suitable for trace analysis.

Q5. Give difference between molar conductance and specific conductance.

Q6. What do you mean by cyclic voltammetry?

Cyclic voltammetry is an electrochemical technique used in chemistry to study the redox behavior of chemical species. It involves applying

a potential voltage to a working electrode and measuring the resulting current as it cycles between two limits. This method provides valuable information about the electron transfer kinetics, electrochemical reversibility, and concentration of species in a solution, making it a crucial tool in electro-analytical chemistry and materials science.

Q7. Give the changes in conductance in the titration of mixture of HCl and acetic acid with NaOH.

In the titration of a mixture of HCl (hydrochloric acid) and acetic acid (CH_3COOH) with NaOH (sodium hydroxide), changes in conductance occur due to the different properties of these acids.

Initial Conductance: At the beginning of the titration, the conductance will primarily be determined by the presence of HCl, which is a strong acid and dissociates almost completely in solution, producing a high concentration of H^+ ions and Cl^- ions. This results in high conductance.

Near Equivalence Point: As you add NaOH to the mixture, it reacts with HCl to form NaCl, a salt. The concentration of H^+ ions decreases, and the conductance starts to decrease. However, acetic acid is a weak acid, and it doesn't dissociate as completely as HCl. So, as the concentration of HCl decreases, the contribution of acetic acid to the conductance becomes more significant.

Equivalence Point: At the equivalence point, all the HCl in the mixture has reacted with NaOH, so the conductance will be primarily determined by acetic acid. Acetic acid will start to dissociate to form

acetate ions (CH_3COO^-) and H^+ ions, leading to an increase in conductance as the acetic acid gets neutralized.

Beyond Equivalence Point: After the equivalence point, the excess NaOH will continue to react with acetic acid, causing an increase in the concentration of acetate ions and a further increase in conductance.

Q8. What is cell constant? What is relation between conductivity of cell and cell constant?

The cell constant is defined as the ratio of distance between the electrodes which is divided by the area of the cross-sectional of the electrode.

Relation

Conductivity of cell is directly proportional to the cell constant.

Conductivity (κ) = $\frac{1}{\text{Resistance (R)}} \times \text{Cell constant}$.

Q9. What different arc and spark sources used in spectroscopy?

Arc Sources	Spark Sources
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• Carbon Arc Lamp • Xenon Arc Lamp • Mercury Arc Lamp • Metal-Halide Arc Lamp • Plasma Arc Cutting	• Spark Erosion • Spark Spectrometry • Sparks in Lightning • Electrical Discharge in Plasmas • Spark Gaps in High-Voltage Testing
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Q10. Give three advantages of DME.

Dimethyl ether (DME) has several advantages in various applications:

- ***Environmentally Friendly:*** DME is a clean-burning fuel that produces minimal emissions, making it an eco-friendly alternative to traditional fossil fuels.
- ***High Energy Density:*** DME has a high energy content per unit volume, making it efficient for transportation and energy storage applications.
- ***Versatility:*** DME can be used as a fuel in engines, a propellant in aerosol products, and a refrigerant, showcasing its versatility in different industries.

LONG QUESTIONS

Q3 (a) Discuss quantitative applications of Anodic stripping voltammetry.

Anodic Stripping Voltammetry (ASV) is a versatile electroanalytical technique that has various quantitative applications, particularly in the determination of trace metals and metalloids in solution. *Some quantitative applications of ASV include:*

Trace Metal Analysis

ASV is widely used for the quantitative determination of trace metals, such as lead, cadmium, copper, mercury, and many others. It can detect these metals at very low concentrations, making it essential in environmental monitoring, food analysis, and industrial quality control.

Environmental Monitoring

ASV is employed for assessing the presence of heavy metals in natural waters, soils, and sediments. This is crucial for understanding environmental contamination and ensuring compliance with regulatory limits.

Quality Control in Electroplating

ASV is utilized in the electroplating industry to monitor and control the metal ion concentration in plating baths. It helps ensure consistent plating quality.

Pharmaceutical Analysis

ASV can be used to determine the concentration of metal impurities in pharmaceutical products, ensuring their safety and compliance with regulatory standards.

Biological and Clinical Analysis

ASV is applied in the analysis of biological samples, including blood, urine, and tissues, to determine metal concentrations for diagnostic and research purposes.

Food and Beverage Testing

ASV is used to assess the metal content in food and beverages, helping to ensure product safety and compliance with food regulations.

Industrial Wastewater Monitoring

ASV is used to assess the metal content in industrial wastewater to ensure compliance with environmental regulations.

Research in Electrochemistry

ASV is widely used in fundamental research related to electrochemistry, providing insights into metal speciation and electrochemical behavior.

Q3 (b) Write a note on conductometric titration of strong acid with NaOH and NH₄OH.

Conductometric titration is a technique used to determine the concentration of a solution by measuring its electrical conductivity.

When performing a conductometric titration of a strong acid with NaOH (sodium hydroxide) and NH_4OH (ammonium hydroxide), you can observe different conductance changes due to the different properties of the titrants.

Titration with NaOH (Sodium Hydroxide)

- Initially, the conductance will be low because the solution contains the strong acid, which fully ionizes, producing a high concentration of H^+ and Cl^- ions.
- As you add NaOH, it reacts with HCl to form water and NaCl, reducing the concentration of H^+ ions and increasing the concentration of Na^+ and Cl^- ions. Conductance starts to increase as the solution becomes less acidic.
- At the equivalence point, all the HCl has reacted with NaOH, and the solution is primarily NaCl, which is a strong electrolyte. Conductance is at its maximum.
- Beyond the equivalence point, excess NaOH is present, and conductance continues to increase as more NaOH is added.

Titration with NH_4OH (Ammonium Hydroxide)

- Initially, the conductance will be low because the solution contains the strong acid, which fully ionizes.
- As you add NH_4OH , it reacts with HCl to form ammonium chloride (NH_4Cl), ammonia (NH_3), and water. The ammonia formed will partially dissociate and increase the concentration of NH_4^+ and OH^- ions, resulting in an increase in conductance.

- At the equivalence point, all the HCl has reacted with NH_4OH , and the solution contains NH_4Cl and NH_3 , which contribute to the conductance.
- Beyond the equivalence point, excess NH_4OH is present, and conductance continues to increase.

Q4 (a) Discuss electrodeposition step and voltammetric completion step in voltammetry.

In voltammetry, which is an electroanalytical technique, two key steps are often involved:

Electrodeposition Step

- ***Purpose:*** The electrodeposition step is the initial stage of a voltammetry experiment and serves the purpose of depositing or accumulating the analyte of interest on the working electrode.
- ***Procedure:*** During this step, the potential at the working electrode is held at a specific value (usually more negative than the standard electrode potential of the analyte) to facilitate the reduction of the analyte species, causing it to accumulate on the electrode surface. The duration of this step may vary based on the experiment and the desired amount of analyte deposition.
- ***Outcome:*** At the end of the electrodeposition step, you have a known quantity of the analyte on the electrode surface, which will be used for the subsequent voltammetric measurements.

Voltammetric Completion Step

- **Purpose:** After the electrodeposition step, the voltammetric completion step is carried out to investigate the electrochemical behavior of the accumulated analyte on the electrode surface.
- **Procedure:** During this step, the applied potential is scanned or swept over a specific range. As the potential changes, the accumulated analyte undergoes oxidation or reduction reactions, generating a current response at the electrode. The current response is recorded as a function of potential.
- **Outcome:** The voltammetric completion step provides valuable information about the redox behavior of the analyte, including its electrochemical reversibility, oxidation/reduction potentials, and concentration. By analyzing the current vs. potential curve (voltammogram), you can obtain quantitative and qualitative information about the analyte.

Q4 (b) Discuss half wave potential. What information is obtained from polarographic curve?

Half wave potential

The half-wave potential, often denoted as $E_{1/2}$, is a key parameter in electrochemistry. It refers to the electrode potential at which half of a redox reaction is complete, specifically, the point at which half of the electroactive species are oxidized (or reduced) at the working electrode. In other words, it's the potential at which the current response in a voltammetric experiment reaches its maximum or

minimum value, depending on whether it's an oxidation or reduction process.

Information obtained from polarographic

Information that can be obtained from a polarographic curve is:

- ***Half-Wave Potential ($E_{1/2}$):*** The potential at which the current reaches its maximum or minimum (depending on whether it's an oxidation or reduction process) is the half-wave potential ($E_{1/2}$). It indicates the electrode potential at which half of the redox reaction is complete. $E_{1/2}$ provides information about the electrochemical reversibility of the reaction and can be used to calculate thermodynamic parameters.
- ***Peak Current (I_p):*** The peak current corresponds to the maximum (or minimum) current observed in the polarographic curve. It is proportional to the concentration of the electroactive species and can be used for quantitative analysis.
- ***Half-Wave Width ($\Delta E_{1/2}$):*** The half-wave width is the potential difference between the anodic and cathodic half-wave potentials ($E_{1/2}$). It provides information about the kinetics of the redox reaction. A narrow half-wave width suggests fast kinetics, while a broad half-wave width indicates slower reaction rates.
- ***Shape of the Curve:*** The shape of the polarographic curve can reveal whether the redox reaction is reversible, quasi-reversible, or irreversible. A symmetric curve with sharp peaks indicates

reversibility, while a skewed or broad curve suggests irreversibility or sluggish kinetics.

- **Redox Mechanism:** The interpretation of polarographic curves can provide insights into the reaction mechanism and pathways for the redox process.

Q5 (a) Write a note on spark sources.

"Spark sources" can refer to different devices or methods used for producing sparks or electrical discharges. There are a few types of spark sources:

- **Spark Erosion (Electric Discharge Machining):** Spark erosion is a machining process that uses electrical discharges (sparks) between an electrode and a work piece to remove material. It is used in the manufacturing industry for cutting, shaping, and drilling hard materials like metals and alloys.
- **Spark Spectrometry:** Spark spectrometry, also known as spark emission spectrometry or spark optical emission spectrometry (Spark-OES), is an analytical technique used for the qualitative and quantitative analysis of metals and alloys.
- **Spark Ignition in Internal Combustion Engines:** In internal combustion engines, spark plugs are used to initiate the combustion process by creating sparks to ignite the air-fuel mixture. These sparks are produced by a spark source within the spark plug.

- ***Sparks in Lightning:*** Natural lightning is essentially a giant spark discharge in the atmosphere, caused by the build-up of electrical charge in thunderstorms. Lightning has applications in atmospheric science and is studied for its electrical and meteorological properties.
- ***Spark Gaps in High-Voltage Testing:*** Spark gaps are used as electrical switches in high-voltage and high-current testing applications. They allow controlled discharges of electrical energy and are used to protect equipment from overvoltage conditions.
- ***Electrical Discharge in Plasmas:*** Spark sources can be used to generate plasmas for various applications, such as plasma cutting and welding, plasma thrusters in spacecraft, and plasma etching in microfabrication processes.

Q5 (b) Write a note on Qualitative aspects of polarographic analysis.

Qualitative aspects in polarographic analysis refer to the information and insights obtained about the electrochemical behavior of analytes in a solution without necessarily quantifying their concentration. Here are some qualitative aspects of polarographic analysis:

Redox Mechanism

The shape of the polarographic curve, particularly the number and positions of peaks, can provide clues about the redox mechanism.

Different reaction pathways may result in distinct polarographic patterns.

Interactions and Complex Formation

Polarography can reveal the presence of complexation reactions or interactions between the analyte and other species in the solution. The formation of complexes can result in changes in peak positions or shapes.

Detection of Impurities

Polarographic analysis may detect the presence of impurities or interfering substances in the sample, which can manifest as additional peaks or disturbances in the polarographic curve.

Identification of Redox Couples

In multi-component systems, polarography can assist in identifying different redox couples and their corresponding reduction or oxidation potentials based on the position of peaks in the curve.

Identification of Unknown Species

In some cases, polarography can be used for qualitative analysis to identify unknown species in a sample. This is achieved by comparing the observed polarographic behavior with known standards or reference compounds.

Solvent and Electrolyte Effects

Qualitative information can be obtained about the influence of the solvent and electrolyte on the redox behavior of the analyte. Changes in the solvent or the type and concentration of the supporting electrolyte may affect the polarographic response.

Characterization of Electrochemical Cells

Qualitative aspects can provide insights into the performance and behavior of electrochemical cells, such as working electrodes and reference electrodes.

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(2022)

SHORT QUESTIONS

Q1. What are three basic mechanisms of mass transport in voltammetry.

- **Diffusion:** The movement of electroactive species due to concentration gradients in the solution.
- **Convection:** Transport driven by bulk fluid flow, such as stirring or flowing solutions.
- **Migration:** Movement of ions in response to an electric field, typically seen in electrolyte .

These mechanisms play crucial roles in the electrochemical processes observed in voltammetry experiments

Q2. Differentiate between cathodic current and anodic current in polarography?

Cathodic Current	Anodic Current
In polarography, cathodic current refers to the flow of electrons from the working electrode to reduce ions or molecules in the solution.	In polarography anodic current represents the flow of electrons in the opposite direction, from the working electrode to oxidize species in the solution,
It occurs at a potential more negative than the formal potential of the species being reduced.	It occurs at potentials more positive than the species' formal potential.
It is associated with the reduction of electroactive species at the electrode surface, where these species gain electrons and are	It is associated with the oxidation of electroactive species at the electrode surface, where these species lose electrons and are

reduced from a higher oxidation state to a lower one.	oxidized from a lower oxidation state to a higher one.
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Q3. Describe the role of inert supporting electrolyte in polarography ?

- In polarography, an inert supporting electrolyte plays a crucial role by providing a stable ionic environment for the electrochemical reaction to occur.
- It helps maintain a constant ionic strength in the solution, preventing changes in pH and conductivity that could affect the accuracy of measurements.
- This supporting electrolyte typically consists of inert salts like potassium chloride (KCl) or sodium sulfate (Na_2SO_4), which do not participate in the electrochemical reaction but enhance the overall conductivity of the solution.
- Ultimately, the inert supporting electrolyte ensures reliable and reproducible results in polarographic experiments by stabilizing the electrochemical conditions.

Q4. How supporting takes place in glow discharge?

Q5. Automation possess some disadvantages. Describe them.

It has several disadvantages:

High Initial Costs: Implementing automation systems can be expensive, requiring significant investments in technology, equipment, and infrastructure.

Job Displacement: Automation often leads to the displacement of human workers, particularly in repetitive or low-skilled tasks. This can result in job loss, unemployment.

Maintenance and Technical Issues: Automated systems require regular maintenance and technical support to function efficiently. Technical failures can disrupt operations and may require specialized skills for troubleshooting and repair.

Complexity and Learning Curve: Automation systems can be complex, and employees need to be trained to operate and maintain them effectively. The learning curve can be steep, and not all employees may adapt easily to new technology.

Q6. Write down the principle of amperometry.

The principle of amperometry is based on measuring the current (amperes) that flows through an electrochemical cell or electrode when a constant voltage is applied. It's commonly used in analytical chemistry to detect and quantify substances, often involving redox reactions. The current is proportional to the concentration of the analyte, following Faraday's laws of electrolysis. This technique is sensitive and allows for precise determination of substances at low concentrations, making it valuable in various applications, including chemical analysis and sensor development.

Q7. How glow discharge atomization take place?

Glow discharge is based on a phenomenon called sputtering. The sample is introduced into a plasma chamber, where a low-pressure inert gas, such as argon, is used to create a discharge. The plasma is generated by applying a high-voltage electrical potential between the cathode and anode within the discharge chamber. This creates a stable glow discharge.

In the plasma, electrons are accelerated, and collisions occur between electrons, argon ions, and the sample introduced into the system.

During these collisions, the high-energy electrons transfer their energy to the atoms or ions of the sample. This process results in the atomization of the sample, converting it into atomic species.

Q8. Distinguish between fixed automation and flexible automation.

Fixed Automation	Flexible Automation
Fixed automation, also known as "hard" or "dedicated" automation, is designed for a specific, repetitive task or a set of tasks.	Flexible automation, often referred to as "soft" or "programmable" automation, is designed to handle various tasks and adapt to different product variations.
It is highly specialized and tailored to a particular production process.	It can be reprogrammed or reconfigured for different production requirements.

Fixed automation systems are inflexible and not easily adaptable to new products or tasks.	Flexible automation systems are versatile and can accommodate changes in product design, production volume, and process requirements.
Fixed automation is typically used for high-volume, mass production.	It is suitable for low-volume, high-mix production scenarios.

Q9. How conductance is measured?

Conductance, which is a measure of the ability of a material to conduct electric current, is typically measured using a conductance meter or a conductance cell.

The conductance meter applies an alternating current (AC) voltage across the electrodes. The current passing through the sample is then measured. The conductance meter typically displays the conductance value in siemens (S) or microsiemens (μS), where $1 \text{ S} = 1,000,000 \mu\text{S}$.

Q10. Write down the basic principle of conductometric titration.

Conductometric titration is a technique used in analytical chemistry to determine the concentration of a specific ion or substance in a solution. The principle behind conductometric titration is based on the measurement of the electrical conductivity of the solution as a reactant is added to it.

Q11. Give disadvantage of anodic stripping voltammetry?

Anodic stripping voltammetry has several disadvantages.

- Firstly, it requires a deposition step, which can be time-consuming and may limit the analysis speed.
- Secondly, it's sensitive to contamination and requires careful electrode preparation and maintenance.
- Thirdly, the technique is mainly applicable to metal ions, limiting its versatility for other types of analytes.
- Lastly, it can be complex and requires skilled operators, making it less accessible for use in some laboratories.

Q12. Distinguish between migration current and residual current?

Migration Current	Residual Current
Migration current refers to the movement of charged ions within a chromatographic column during the separation process.	Residual current relates to the small, lingering electrical current in a circuit even when no intentional current is flowing.
It is a dynamic process that helps separate components based on their affinity for the stationary phase.	It is often associated with electrical leakage or imbalance and can be a safety concern.

Q13. Describe amperometric titration involving electro reducible vs electro oxidiseable species.

In amperometric titration, electro reducible and electro oxidizable species refer to the types of analytes being measured using an electrochemical method. *The difference between the two is:*

Amperometric Titration Involving Electro-Reducible Species

- Electro reducible species are substances that can undergo reduction reactions at the working electrode of the electrochemical cell during the titration.
- Reduction involves gaining electrons and often corresponds to a decrease in the oxidation state of the species.
- In amperometric titration, a reducing agent is added to the solution, and the current (amperage) generated due to the reduction of the electro reducible species is measured. This current is used to determine the analyte's concentration.
- **Example:** A common electro reducible species is iodine (I_2), which can be reduced to iodide ions (I^-) during titration.

Amperometric Titration Involving Electro-Oxidizable Species

- Electro oxidizable species are substances that can undergo oxidation reactions at the working electrode of the electrochemical cell during the titration.
- Oxidation involves losing electrons and often corresponds to an increase in the oxidation state of the species.

- In amperometric titration, an oxidizing agent is added to the solution, and the current (amperage) generated due to the oxidation of the electro oxidizable species is measured. This current is used to determine the analyte's concentration.
- **Example:** A common electro oxidizable species is ascorbic acid (vitamin C), which can be oxidized to dehydroascorbic acid during titration.

Q14. What is difference between arc and spark?

Q15. What is the effect of charge and size of ions on conductivity?

The charge and size of ions in a solution have a significant impact on the electrical conductivity of that solution.

Effect of Ion Charge

- Ions with a higher electric charge contribute to higher electrical conductivity. This is because ions with greater charge carry more electrical charge, and they are more effective at carrying an electric current through the solution.
- The concentration of ions in the solution is crucial. Even if ions have high charges, if their concentration is low, the overall conductivity will be reduced.

Effect of Ion Size

- The size of ions can affect their mobility in a solution. Smaller ions tend to move more rapidly than larger ions, which can impact their contribution to electrical conductivity.

- Smaller ions can move through the solution more easily and thus contribute more to electrical conductivity compared to larger ions.

LONG QUESTIONS

Q2(i) Discuss arc and spark sources.

Arc sources

Arc sources are devices used for producing intense light or plasma by creating an electric arc. These sources are commonly used in various applications, including lighting, welding, spectroscopy, and materials processing. *Here are some common types of arc sources:*

Carbon Arc Lamp: A carbon arc lamp consists of two carbon rods or electrodes. An electric current is passed between the electrodes, creating a high-temperature electric arc. These lamps produce a bright, intense light used for street lighting, projectors, and film production.

Xenon Arc Lamp: Xenon arc lamps use xenon gas as the arc medium. They are widely used in cinema projectors, automotive headlights, and some types of scientific instruments. Xenon arc lamps produce a continuous spectrum of light, making them suitable for applications where color accuracy is important.

Mercury Arc Lamp: Mercury arc lamps contain mercury vapor and produce intense ultraviolet (UV) light when an arc is struck. They are

commonly used in UV spectroscopy, fluorescence microscopy, and other applications that require UV light sources.

Metal-Halide Arc Lamp: Metal-halide arc lamps are similar to mercury arc lamps but contain metal halides in addition to mercury vapor. These lamps produce a broader spectrum of light, making them suitable for applications like stadium lighting and stage lighting.

Plasma Arc Cutting: Plasma arc cutting is a process that uses a high-temperature plasma arc to cut through metals. It involves passing a gas (commonly argon) through a nozzle and applying an electric arc to create a plasma jet that melts and removes the metal.

Spectroscopic Arc Sources: Arc sources are also used in atomic emission spectroscopy (AES) and atomic absorption spectroscopy (AAS). These sources create a high-temperature plasma in which atoms are excited and emit characteristic wavelengths of light for elemental analysis.

Spark Sources

"Spark sources" can refer to different devices or methods used for producing sparks or electrical discharges. There are a few types of spark sources:

Spark Erosion (Electric Discharge Machining): Spark erosion is a machining process that uses electrical discharges (sparks) between an electrode and a work piece to remove material. It is used in the

manufacturing industry for cutting, shaping, and drilling hard materials like metals and alloys.

Spark Spectrometry: Spark spectrometry, also known as spark emission spectrometry or spark optical emission spectrometry (Spark-OES), is an analytical technique used for the qualitative and quantitative analysis of metals and alloys.

Spark Ignition in Internal Combustion Engines: In internal combustion engines, spark plugs are used to initiate the combustion process by creating sparks to ignite the air-fuel mixture. These sparks are produced by a spark source within the spark plug.

Sparks in Lightning: Natural lightning is essentially a giant spark discharge in the atmosphere, caused by the build-up of electrical charge in thunderstorms. Lightning has applications in atmospheric science and is studied for its electrical and meteorological properties.

Spark Gaps in High-Voltage Testing: Spark gaps are used as electrical switches in high-voltage and high-current testing applications. They allow controlled discharges of electrical energy and are used to protect equipment from overvoltage conditions.

Electrical Discharge in Plasmas: Spark sources can be used to generate plasmas for various applications, such as plasma cutting and welding, plasma thrusters in spacecraft, and plasma etching in microfabrication processes.

Q2 (ii) Explain how conductometric titrations are applied in complexometric and participations reactions

Conductometric titrations are a type of titration in which the electrical conductivity of a solution is measured as a function of the volume of titrant added. Conductometric titrations are commonly used in complexometric and precipitation reactions for determining the concentration of various analytes. Here's how they are applied in these types of reactions:

Complexometric Titrations

- ***Complex Formation:*** In complexometric titrations, a complexing agent (often a chelating agent) is added to the sample solution containing the analyte. The complexing agent forms a stable complex with the analyte, resulting in a decrease in electrical conductivity due to a decrease in the concentration of free ions in the solution.
- ***Titration Process:*** A standard solution of a titrant, which can also form a complex with the analyte, is slowly added to the sample solution. As the titrant reacts with the analyte, the concentration of the analyte complex decreases, and the conductivity of the solution gradually increases.
- ***Endpoint Detection:*** The endpoint of the titration is determined by monitoring the change in electrical conductivity. It's often recognized as a sudden and significant increase in conductivity, indicating that the analyte complex has been fully titrated.

- **Calculation:** The concentration of the analyte can be calculated using the volume and concentration of the titrant, as well as the stoichiometry of the complexation reaction.
- **Uses:** Complexometric titrations are commonly used for the determination of metal ions, especially those that form stable complexes with chelating agents. EDTA (ethylene diamine tetra acetic acid) is a widely used complexing agent in these titrations.

Precipitation Titrations

- **Formation of Precipitate:** In precipitation titrations, a reagent is added to the sample solution, causing the analyte to form an insoluble precipitate. This leads to a decrease in electrical conductivity as the precipitate settles.
- **Titration Process:** A titrant solution, which reacts with the precipitate to form a soluble product, is carefully added to the sample solution. This process converts the precipitate back into a soluble form, resulting in an increase in electrical conductivity.
- **Endpoint Detection:** The endpoint is typically recognized by monitoring the change in electrical conductivity. As the titration nears completion, a significant increase in conductivity indicates that all of the precipitate has dissolved.
- **Calculation:** The concentration of the analyte can be calculated using the volume and concentration of the titrant, along with the stoichiometry of the reaction.

- **Uses:** Precipitation titrations are commonly used for the determination of anions or cations that can form insoluble salts or precipitates with specific reagents.

Q2 (iii) Discuss instrumentation in amperometric titrations.

Amperometric titrations are a type of analytical technique used to determine the concentration of a specific substance in a solution. In these titrations, an amperometric sensor is a crucial component. The instrumentation typically includes:

1. Electrode

An amperometric titration setup usually consists of a working electrode, which is sensitive to the analyte of interest. The electrode is often made of materials like platinum, gold, or glassy carbon.

2. Reference Electrode

A reference electrode is used to provide a stable reference potential. Common reference electrodes include the silver/silver chloride electrode (Ag/AgCl) or a saturated calomel electrode (SCE).

3. Counter Electrode

This electrode completes the electrical circuit and ensures the flow of current. It's often made of an inert material like platinum.

4. Potentiostat/Galvanostat

This instrument controls the potential (voltage) applied to the working electrode and measures the resulting current. It allows for precise control of the electrochemical conditions during the titration.

5. Analyte Solution

The solution containing the analyte to be determined is titrated with a titrant of known concentration.

6. Titrant

The titrant is added incrementally to the analyte solution. The change in current at the working electrode is recorded as the titration progresses.

7. Data Acquisition System

A system for collecting and recording the current values as a function of the volume of titrant added.

8. Software

Software is used to control the titration process, record data, and analyze the results.

Q2(iv) describe the working and construction of voltammetric cell?

A voltammetric cell is an essential component in electrochemical analysis, often used in fields like chemistry. It allows for the

measurement of current as a function of applied voltage, providing valuable information about the redox properties of substances.

Working

Electrodes: A voltammetric cell typically consists of two electrodes: a working electrode and a reference electrode. The working electrode is where the electrochemical reaction of interest occurs, while the reference electrode maintains a stable potential against which the working electrode voltage is measured.

Electrochemical Reaction: The substance to be analyzed, known as the analyte, is introduced into the cell's solution. An applied voltage is then gradually increased or decreased while monitoring the resulting current. As the voltage changes, the electrochemical reaction at the working electrode's surface also changes.

Current Response: The current passing between the working and reference electrodes is directly proportional to the rate of the electrochemical reaction. By plotting the current against the applied voltage (voltammogram), researchers can analyze the substance's redox properties, such as its oxidation and reduction potentials.

Construction

Working Electrode: The working electrode is typically made from a material of interest, often a noble metal like platinum, gold, or glassy carbon. It serves as the site of the electrochemical reaction.

Reference Electrode: The reference electrode is crucial for maintaining a stable potential. Common reference electrodes include the silver/silver chloride electrode (Ag/AgCl) and the saturated calomel electrode (SCE). They are connected to the working electrode through an electrolyte solution.

Electrolyte Solution: The cell contains an electrolyte solution that provides a medium for the ionic movement necessary for the electrochemical reactions. The composition of this solution depends on the specific analysis being conducted.

Cell Housing: The cell components are enclosed within a housing made of chemically resistant materials like glass or plastic to prevent contamination and maintain a controlled environment.

Electrical Circuit: An external potentiostat or galvanostat is used to apply the controlled voltage and measure the resulting current. This instrument provides precise control over the voltage and records the current data.

Q2 (v) Discuss in detail the factors affecting the limiting current?

Limiting current, in the context of electrochemistry, refers to the maximum current that can be sustained during an electrochemical reaction. Several factors affect the limiting current discussed below:

Electrode Surface Area

The most significant factor is the surface area of the electrodes. A larger surface area allows more reactants to come into contact with

the electrode, increasing the rate of reaction and, therefore, the limiting current.

Concentration of Electrolyte

The concentration of the electrolyte in the solution can have a profound effect on the limiting current. Higher concentrations provide more ions for the electrochemical reaction, leading to a higher limiting current.

Temperature

Temperature plays a crucial role in electrochemical reactions. Higher temperatures generally increase the kinetic energy of molecules, leading to faster reaction rates and higher limiting currents.

Electrolyte Viscosity

The viscosity of the electrolyte can impede the movement of ions to the electrode surface. Higher viscosity can hinder ion transport and reduce the limiting current.

Electrode Shape and Structure

The shape and structure of the electrode can impact the limiting current. Rough or porous electrodes provide more surface area for reaction, increasing the limiting current.

Mass Transport

Factors affecting the transport of reactants to the electrode surface, such as diffusion rates and flow rates, can also influence the limiting current.

Presence of Inhibitors or Catalysts

The presence of inhibitors or catalysts in the electrolyte can either decrease or increase the limiting current, respectively, by affecting the reaction kinetics.

Q2 (vi) Polarography is applied to organic and inorganic substances. Explain in detail.

Definition

Polarography is a technique used in analytical chemistry to study and quantify the concentration of certain chemical substances, particularly those that undergo redox (reduction-oxidation) reactions. It is applicable to both organic and inorganic substances and has been widely used in various fields of chemistry.

Principle of Polarography

Polarography relies on the principle of measuring the current passing through an electrochemical cell as a function of the applied potential (voltage) between a working electrode and a reference electrode. The working electrode is typically made of mercury, which forms a liquid

amalgam with other metals and allows for controlled electrodeposition and stripping of analytes.

Applications to Organic Substances

- ***Determination of Organic Compounds:*** Polarography can be used to determine the concentration of organic compounds that are capable of undergoing redox reactions. For example, it's used in the analysis of organic molecules like vitamins, pharmaceuticals, and organic dyes.
- ***Analyzing Organic Reaction Kinetics:*** Polarography is also used to study the kinetics of organic reactions, particularly those involving electron transfer processes. It helps in understanding reaction mechanisms and reaction rates.

Applications to Inorganic Substances

- ***Determination of Inorganic Ions:*** Polarography is widely used for the determination of various inorganic ions in solution, including heavy metals like lead, cadmium, and copper. It is a valuable tool in environmental and industrial analysis to monitor the levels of these ions.
- ***Quality Control in Electroplating:*** In the electroplating industry, polarography is used to control the quality of metal coatings by monitoring the concentration of metal ions in the plating bath.

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